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Tarp rolling device

Abstract

A tarp rolling device includes a handle bracket which includes a first section slidably engaging a second section such that the handle bracket has an adjustable width. A pair of ratcheting spools is removably mountable to the handle bracket and each of the ratcheting spools includes a plurality of rods which extend across a tarp to facilitate the tarp to be rolled into a roll when the pair of ratcheting spools are rotated. The handle bracket is urged to swing downwardly thereby facilitating the handle bracket to be rotated on each of the ratcheting spools. Continuing, the handle bracket is urged to swing upwardly subsequent to being urged downwardly thereby rotating each of the ratcheting spools to wrap the tarp around the plurality of rods.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) Not Applicable

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

(2) Not Applicable

THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT

(3) Not Applicable

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC OR AS A TEXT FILE VIA THE OFFICE ELECTRONIC FILING SYSTEM

(4) Not Applicable

STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR JOINT INVENTOR

(5) Not Applicable

BACKGROUND OF THE INVENTION

(1) Field of the Invention

(6) The disclosure relates to rolling devices and more particularly pertains to a new rolling device

for rolling a tarp into a roll for storage. The device includes a pair of ratcheting spools and a handle bracket that is attachable to the ratcheting spools. Each of the ratcheting spools includes a plurality of rods that extend partially across a tarp. The handle bracket is alternatively pivoted upwardly and downwardly to facilitate the ratcheting spools to roll the tarp around the rods.

(2) Description of Related Art Including Information Disclosed Under 37 CFR 1.97 and 1.98

(7) The prior art relates to rolling devices including a variety of tarp rolling devices that are each attached to a cargo trailer to facilitate a tarp to be either rolled for storage or unrolled to cover the cargo trailer. In no instance does the prior art disclose a portable tarp roller that can roll a tarp which is lying on the ground into a roll for storage.

BRIEF SUMMARY OF THE INVENTION

(8) An embodiment of the disclosure meets the needs presented above by generally comprising a handle bracket which includes a first section slidably engaging a second section such that the handle bracket has an adjustable width. A pair of ratcheting spools is removably mountable to the handle bracket and each of the ratcheting spools includes a plurality of rods which extend across a tarp to facilitate the tarp to be rolled into a roll when the pair of ratcheting spools are rotated. The handle bracket is urged to swing downwardly thereby facilitating the handle bracket to be rotated on each of the ratcheting spools. Continuing, the handle bracket is urged to swing upwardly subsequent to being urged downwardly thereby rotating each of the ratcheting spools to wrap the tarp around the plurality of rods.

(9) There has thus been outlined, rather broadly, the more important features of the disclosure in order that the detailed description thereof that follows may be better understood, and in order that the present contribution to the art may be better appreciated. There are additional features of the disclosure that will be described hereinafter and which will form the subject matter of the claims appended hereto.

(10) The objects of the disclosure, along with the various features of novelty which characterize the disclosure, are pointed out with particularity in the claims annexed to and forming a part of this disclosure.

Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWING(S)

(1) The disclosure will be better understood and objects other than those set forth above will become apparent when consideration is given to the following detailed description thereof. Such description makes reference to the annexed drawings wherein:

(2) FIG. 1 is a top perspective view of a tarp rolling device according to an embodiment of the disclosure.

(3) FIG. 2 is an exploded perspective view of an embodiment of the disclosure.

(4) FIG. 3 is a front view of an embodiment of the disclosure.

(5) FIG. 4 is a right side phantom view of an embodiment of the disclosure.

(6) FIG. 5 is a perspective in-use view of an embodiment of the disclosure.

(7) FIG. 6 is a magnified detail view taken from circle 6 of FIG. 4 of an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

(8) With reference now to the drawings, and in particular to FIGS. 1 through 6 thereof, a new rolling device embodying the principles and concepts of an embodiment of the disclosure and generally designated by the reference numeral 10 will be described.

(9) As best illustrated in FIGS. 1 through 6, the tarp rolling device 10 generally comprises a handle bracket 12 which includes a first section 14 which slidably engages a second section 16 such that the handle bracket 12 has an adjustable width. The first section 14 comprises a first tube 18 that is

perpendicularly oriented with a second tube **20** and the second section **16** comprises a primary tube **15** that is perpendicularly oriented with a secondary tube **17**. The first tube **18** has dimensions that are greater than dimensions of the primary tube **15**. In this way the first tube **18** can insertably receive the primary tube **15** having the second tube **20** being spaced from and being oriented parallel to the secondary tube **17** such that the handle bracket **12** defines a U-shape.

(10) A screw **22** is provided which has a shank **24** that is threadable through a nut **26** which is mounted to the first tube **18** of the first section **14**. The shank **24** compresses against the first tube **18** of the second section **16** when the screw **22** is tightened for retaining the second section **16** at a desired position with respect to the first section **14**. Conversely, the shank **24** is displaced from the first tube **18** of the second section **16** when the screw **22** is loosened to facilitate the second section **16** to be adjusted in the first section **14**. The screw **22** may include a handle **27** that is perpendicularly oriented with the shank **24** for gripping and rotating the screw **22**.

(11) A pair of ratcheting spools **28** is provided and each of the pair of ratcheting spools **28** is removably mountable to the handle bracket **12** such that the pair of ratcheting spools **28** are oriented to face each other. Each of the pair of ratcheting spools **28** includes a wheel **30** which is rotated when the ratcheting spool **28** to which the wheel **30** is attached is rotated. Additionally, each of the ratcheting spools **28** includes a plurality of rods **32** each disposed on the wheel **30** of a respective ratcheting spool **28**. The plurality of rods **32** of each of the ratcheting spools **28** extends partially across a tarp **34** to facilitate the tarp **34** to be rolled into a roll when the pair of ratcheting spools **28** are rotated. The tarp **34** may be a tarp of any conventional size or design that is commonly employed in the trucking industry for covering or securing a load.

(12) Each of the ratcheting spools **28** comprises a spool bracket **36** that has a central panel **38** extending between a pair of outward panels **40** having the pair of outward panels **40** being spaced apart from each other and being oriented parallel to each other. Each of the ratcheting spools **28** includes a member **42** that has an outer surface **44** which is attached to a leading edge **46** of a respective one of the pair of outward panels **40** of the spool bracket **36**. The member **42** extends upwardly beyond the central panel **38** such that an upper end **48** of the member **42** is spaced from the central panel **38**. The second tube **20** the first section **14** and the secondary tube **17** of the second section **16** of the handle bracket **12** insertably receives the member **42** of a respective one of the ratcheting spools **28**.

(13) Each of the ratcheting spools **28** includes a shaft **50** that extends through each of the outward panels **40** such that the shaft **50** is freely rotatable in the spool bracket **36**. The shaft **50** has a first end **52** that is spaced from a first one of the outward panels **54** and the shaft **50** has a second end **56** is spaced from a second one of the outward panels **58**. The wheel **30** has a rear surface **60** that is attached to the second end **56** of the shaft **50** and the wheel **30** has a perimeter edge **62** which alternatively curves toward and away from a center of the wheel **30** to define a plurality of lobes **64** which are evenly spaced apart from each other and distributed around the wheel **30**.

(14) Each of the plurality of rods **32** has a coupled end **66** that is coupled to a front surface **68** of the wheel **30** and each of the rods **32** is located proximate a distal end **70** of a respective one of the lobes **64** defined on the wheel **30**. Each of the rods **32** on the wheel **30** of each of the pair of ratcheting spools **28** is directed toward each other when the pair of ratcheting spools **28** is mounted to the handle bracket **12**. Additionally, the plurality of rods **32** includes a set of first rods **72** and a set of second rods **74** and each of the first rods **72** has a length that is greater than a length of the second rods **74**.

(15) Each of the ratcheting spools **28** includes a gear **76** that is positioned around the shaft **50**. The gear **76** is positioned between the first outward panel **54** and the first end **52** of the shaft **50** and the gear **76** has a plurality of teeth **78** distributed around the gear **76**. Each of the plurality of teeth **78** has a first edge **80** that is perpendicularly oriented with respect to an outer surface **81** of the shaft **50**. Additionally, each of the plurality of teeth **78** has a second edge **82** that is sloped with respect to the first edge **80**.

(16) Each of the ratcheting spools **28** includes a ratchet **84** that is pivotally coupled to an outwardly facing surface **86** of the first outward panel **54**. The ratchet **84** is biased into a locking position having a finger **88** of the ratchet **84** engaging the gear **76**. The finger **88** slides over the second edge **82** of each of the teeth **78** when the spool bracket **36** is rotated about the shaft **50** in a first direction. The finger **88** abuts the first edge **80** of a respective one of the teeth **78** when the finger **88** is in the locked position and the spool bracket **36** is urged to rotate about the shaft **50** in a second direction. In this way the spool bracket **36** is inhibited from rotating about the shaft **50** in the second direction such that the shaft **50** rotates with the spool bracket **36**. The ratchet **84** is urgeable into an unlocked position having the finger **88** being displaced from the gear **76** thereby facilitating the spool bracket **36** to be rotated about the shaft **50** in either the first direction or the second direction.

(17) The first tube **18** of the first section **14** and the first tube **18** of the second section **16** of the handle bracket **12** is urged to swing downwardly thereby rotating the spool bracket **36** of each of the ratcheting spools **28** to rotated about the shaft **50** in the first direction. The finger **88** of the ratchet **84** slides over the second edge **82** of each of the plurality of teeth **78** on the gear **76** thereby inhibiting the shaft **50** from being rotated with the spool bracket **36**. The first tube **18** of the first section **14** and the first tube **18** of the second section **16** is urged to swing upwardly subsequent to being urged downwardly thereby rotating the spool bracket **36** of each of the ratcheting spools **28** to be rotated in the second direction about the shaft **50**. Furthermore, the finger **88** of the ratchet **84** engages the first edge **80** of a respective tooth **78** on the gear **76** thereby facilitating the shaft **50** to be rotated with the spool bracket **36**. In this way each of the ratcheting spools **28** can wrap the tarp **34** around the plurality of rods **32** each time the spool bracket **36** of each of the ratcheting spools **28** is urged to rotate in the second direction.

(18) In use, the first tube **18** of the second section **16** of the handle bracket **12** is inserted into the first tube **18** of the first section **14** of and the member **42** of each of the ratcheting spools **28** is inserted into the second tube **20** of a respective first section **14** and second section **16**. The tarp **34** is stretched out onto the ground, the tarp **34** is folded if necessary and the tarp **34** is positioned to facilitate the rods **32** on each of the ratcheting spools **28** to extend partially across the tarp **34**. The first tube **18** of the first second and the first tube **18** of the second section **16** are repeatedly urged toward the ground and subsequently back up again in order to rotate the wheel **30** on each of the ratcheting spools **28** to roll the tarp **34** around the rods **32** until the tarp **34** is fully rolled up around the rods **32**. In this way a tarp **34** of any size can be easily rolled up by a single individual while the tarp **34** is laying on the ground. The second tube **20** of the first section **14** and the secondary tube **17** of the second section **16** are each lifted from the member **42** on the respective ratcheting spool **28** and each of the ratcheting spools **28** are pulled away from the tarp **34** to pull the rods **32** out from the rolled up tarp **34**. Finally, bungee cords or the like are wrapped around the rolled up tarp **34** to facilitate the tarp **34** to be stored and the pair of ratcheting spools **28** and each of the first section **14** and second section **16** are stored.

(19) With respect to the above description then, it is to be realized that the optimum dimensional relationships for the parts of an embodiment enabled by the disclosure, to include variations in size, materials, shape, form, function and manner of operation, device and use, are deemed readily apparent and obvious to one skilled in the art, and all equivalent relationships to those illustrated in the drawings and described in the specification are intended to be encompassed by an embodiment of the disclosure.

(20) Therefore, the foregoing is considered as illustrative only of the principles of the disclosure. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the disclosure to the exact construction and operation shown and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the disclosure. In this patent document, the word “comprising” is used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded. A reference to an element by the indefinite article “a” does not exclude the possibility

that more than one of the element is present, unless the context clearly requires that there be only one of the elements.

Claims

1. A tarp rolling device for rolling a tarp into a roll for storage, said device comprising: a handle bracket comprising a first section slidably engaging a second section such that said handle bracket has an adjustable width; a pair of ratcheting spools, each of said pair of ratcheting spools being removably mountable to said handle bracket such that said pair of ratcheting spools are oriented to face each other, each of said pair of ratcheting spools including a wheel which is rotated when ratcheting spool to which said wheel is attached is rotated, each of said ratcheting spools including a plurality of rods each being disposed on said wheel of a respective ratcheting spool wherein said plurality of rods of each of said ratcheting spools is configured to extend partially across a tarp to facilitate the tarp to be rolled into a roll when said pair of ratcheting spools are rotated; wherein said handle bracket is urged to swing downwardly thereby facilitating said handle bracket to be rotated on each of said ratcheting spools; and wherein said handle bracket is urged to swing upwardly subsequent to being urged downwardly thereby rotating wherein each of said ratcheting spools is configured to wrap the tarp around said plurality of rods each time said ratcheting spools is urged to rotate.
2. The device according to claim 1, wherein said first section comprises a first tube being perpendicularly oriented with a second tube, said second section comprises a primary tube being perpendicularly oriented with a secondary tube, said first tube having dimensions being greater than dimensions of said primary tube thereby facilitating said first tube to insertably receive said primary tube said second tube being spaced from and being oriented parallel to said secondary tube such that said handle bracket defines a U-shape.
3. The device according to claim 2, further comprising a screw having a shank being threadable through a nut being mounted to said first tube of said first section, said shank compressing against said first tube of said second section when said screw is tightened for retaining said second section at a desired position with respect to said first section, said shank being displaced from said first tube of said second section when said screw is loosened to facilitate said second section to be adjusted in said first section.
4. The device according to claim 2, wherein each of said ratcheting spools comprises: a spool bracket having a central panel extending between a pair of outward panels having said pair of outward panels being spaced apart from each other and being oriented parallel to each other; and a member having an outer surface being attached to a leading edge of a respective one of said pair of outward panels of said spool bracket, said member extending upwardly beyond said central panel such that an upper end of said member is spaced from said central panel, said second tube of said first section and said secondary tube of said second section of said handle bracket insertably receiving said member of a respective one of said ratcheting spools.
5. The device according to claim 4, wherein: each of said ratcheting spools includes a shaft extending through each of said outward panels such that said shaft is freely rotatable in said spool bracket; said shaft has a first end being spaced from a first one of said outward panels; said shaft has a second end being spaced from a second one of said outward panels; said wheel has a rear surface being attached to said second end of said shaft; and said wheel has a perimeter edge which alternatively curves toward and away from a center of said wheel to define a plurality of lobes being evenly spaced apart from each other and being distributed around said wheel.
6. The device according to claim 5, wherein: each of said plurality of rods has a coupled end being coupled to a front surface of said wheel; each of said rods is located proximate a distal end of a respective one of said lobes defined on said wheel; each of said rods on said wheel of each of said pair of ratcheting spools is directed toward each other when said pair of ratcheting spools is

mounted to said handle bracket; said plurality of rods includes a set of first rods and a set of second rods; and each of said first rods has a length being greater than a length of said second rods.

7. The device according to claim 5, wherein: each of said ratcheting spools includes a gear being positioned around said shaft; said gear is positioned between said first outward panel and said first end of said shaft; said gear has a plurality of teeth being distributed around said gear; each of said plurality of teeth has a first edge being perpendicularly oriented with respect to an outer surface of said shaft; and each of said plurality of teeth has a second edge being sloped with respect to said first edge.

8. The device according to claim 7, wherein: each of said ratcheting spools includes a ratchet being pivotally coupled to an outwardly facing surface of said first outward panel; said ratchet being biased into a locking position having a finger of said ratchet engaging said gear; said finger sliding over said second edge of each of said teeth when said spool bracket is rotated about said shaft in a first direction; said finger abuts said first edge of a respective one of said teeth when said finger is in said locked position and said spool bracket is urged to rotate about said shaft in a second direction thereby inhibiting said spool bracket from rotating in about said shaft in said second direction such that said shaft rotates with said spool bracket; and said ratchet is urgeable into an unlocked position having said finger being displaced from said gear thereby facilitating said spool bracket to be rotated about said shaft in either said first direction or said second direction.

9. The device according to claim 8, wherein said first tube of said first section and said primary tube of said second section of said handle bracket is urged to swing downwardly thereby rotating said spool bracket of each of said ratcheting spools to rotated about said shaft in said first direction having said finger of said ratchet sliding over said second edge of each of said plurality of teeth on said gear thereby inhibiting said shaft from being rotated with said spool bracket.

10. The device according to claim 9, wherein said first tube of said first section and said primary tube of said second section is urged to swing upwardly subsequent to being urged downwardly thereby rotating said spool bracket of each of said ratcheting spools to be rotated in said second direction about said shaft having said finger of said ratchet engaging said first edge of a respective tooth on said gear thereby facilitating said shaft to be rotated with said spool bracket wherein each of said ratcheting spools is configured to wrap the tarp around said plurality of rods each time said spool brackets of each of said ratcheting spools is urged to rotate in said second direction.

11. A tarp rolling device for rolling a tarp into a roll for storage, said device comprising: a handle bracket comprising a first section slidably engaging a second section such that said handle bracket has an adjustable width said first section comprises a first tube being perpendicularly oriented with a second tube, said second section comprises a primary tube being perpendicularly oriented with a secondary tube, said first tube having dimensions being greater than dimensions of said primary tube thereby facilitating said first tube to insertably receive said primary tube said second tube being spaced from and being oriented parallel to said secondary tube such that said handle bracket defines a U-shape; a screw having a shank being threadable through a nut being mounted to said first tube of said first section, said shank compressing against said first tube of said second section when said screw is tightened for retaining said second section at a desired position with respect to said first section, said shank being displaced from said first tube of said second section when said screw is loosened to facilitate said second section to be adjusted in said first section; a pair of ratcheting spools, each of said pair of ratcheting spools being removably mountable to said handle bracket such that said pair of ratcheting spools are oriented to face each other, each of said pair of ratcheting spools including a wheel which is rotated when ratcheting spool to which said wheel is attached is rotated, each of said ratcheting spools including a plurality of rods each being disposed on said wheel of a respective ratcheting spool wherein said plurality of rods of each of said ratcheting spools is configured to extend partially across a tarp to facilitate the tarp to be rolled into a roll when said pair of ratcheting spools are rotated, each of said ratcheting spools comprising: a spool bracket having a central panel extending between a pair of outward panels having said pair of

outward panels being spaced apart from each other and being oriented parallel to each other; a member having an outer surface being attached to a leading edge of a respective one of said pair of outward panels of said spool bracket, said member extending upwardly beyond said central panel such that an upper end of said member is spaced from said central panel, said second tube of said first section and said secondary tube of said second section of said handle bracket insertably receiving said member of a respective one of said ratcheting spools; a shaft extending through each of said outward panels such that said shaft is freely rotatable in said spool bracket, said shaft having a first end being spaced from a first one of said outward panels, said shaft having a second end being spaced from a second one of said outward panels, said wheel having a rear surface being attached to said second end of said shaft, said wheel having a perimeter edge which alternatively curves toward and away from a center of said wheel to define a plurality of lobes being evenly spaced apart from each other and being distributed around said wheel, each of said plurality of rods has a coupled end being coupled to a front surface of said wheel, each of said rods being located proximate a distal end of a respective one of said lobes defined on said wheel, each of said rods on said wheel of each of said pair of ratcheting spools being directed toward each other when said pair of ratcheting spools is mounted to said handle bracket, said plurality of rods including a set of first rods and a set of second rods, each of said first rods has a length being greater than a length of said second rods; a gear being positioned around said shaft, said gear being positioned between said first outward panel and said first end of said shaft, said gear having a plurality of teeth being distributed around said gear, each of said plurality of teeth having a first edge being perpendicularly oriented with respect to an outer surface of said shaft, each of said plurality of teeth having a second edge being sloped with respect to said first edge; and a ratchet being pivotally coupled to an outwardly facing surface of said first outward panel, said ratchet being biased into a locking position having a finger of said ratchet engaging said gear, said finger sliding over said second edge of each of said teeth when said spool bracket is rotated about said shaft in a first direction, said finger abutting said first edge of a respective one of said teeth when said finger is in said locked position and said spool bracket is urged to rotate about said shaft in a second direction thereby inhibiting said spool bracket from rotating in about said shaft in said second direction such that said shaft rotates with said spool bracket, said ratchet being urgeable into an unlocked position having said finger being displaced from said gear thereby facilitating said spool bracket to be rotated about said shaft in either said first direction or said second direction; wherein said first tube of said first section and said primary tube of said second section of said handle bracket is urged to swing downwardly thereby rotating said spool bracket of each of said ratcheting spools to rotated about said shaft in said first direction having said finger of said ratchet sliding over said second edge of each of said plurality of teeth on said gear thereby inhibiting said shaft from being rotated with said spool bracket; and wherein said first tube of said first section and said primary tube of said second section is urged to swing upwardly subsequent to being urged downwardly thereby rotating said spool bracket of each of said ratcheting spools to be rotated in said second direction about said shaft having said finger of said ratchet engaging said first edge of a respective tooth on said gear thereby facilitating said shaft to be rotated with said spool bracket wherein each of said ratcheting spools is configured to wrap the tarp around said plurality of rods each time said spool brackets of each of said ratcheting spools is urged to rotate in said second direction.
